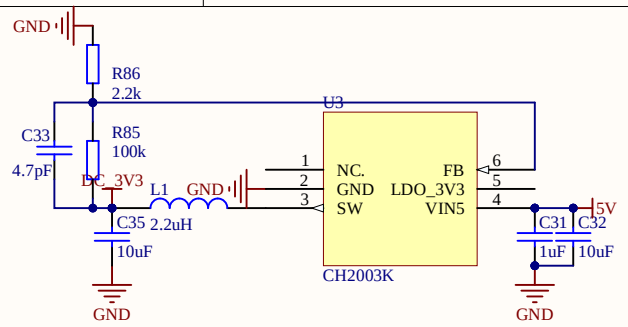
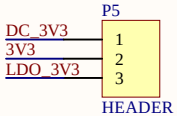

POWER
POWER.SchDoc
5V
USB_LINK
USBPD
USB3.0
3V3
1V2
OTG
Link
LINK.SchDoc
5V
USB_LINK
LINK
3V3
QSPI
QSPI.SchDoc
QSPI
3V3
SDRAM
SDRAM.SchDoc
SDRAM
3V3
LCD
LCD.SchDoc
5V
LCD_LTDC
TOUCH_INT
TOUCH_RST
LCD_TOUCH_IIC
3V3
DVP
DVP.SchDoc
DVP_RST
DVP_PWDN
DVP
3V3
U_MCU
IOS.SchDoc
USBPD
USB3.0
3V3
1V2
OTG
LINK
QSPI
SDRAM
LTDC
TOUCH_INT
TOUCH_RST
IIC
DVP_RST
DVP_PWDN
DVP
VIO_1.8
SDMMC
ETH
LED0
LED1
SAI
PIOC
SERDES
HSADC
I3C
DFSDM
CAN
eMMC
eMMC.SchDoc
VIO_1.8
eMMC
3V3
ETH
ETH.SchDoc
3V3
ETH
LED0
LED1
SAI
SAI.SchDoc
SAI
3V3
GPIO_OUT
GPIO_OUT.SchDoc
PIOC
SERDES
HSADC
I3C
I3C.SchDoc
I3C
3V3
DFSDM
DFSDM.SchDoc
DFSDM
3V3
CAN
CAN.SchDoc
CAN
5V

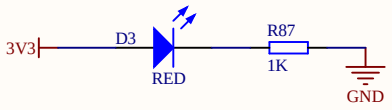
3V3 NET

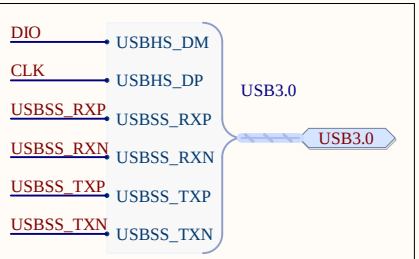


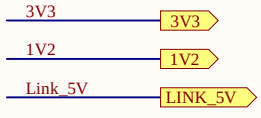
POWER SEL



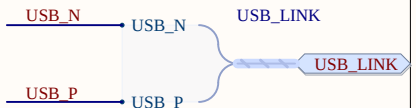
Default bypass cap LDO_3V3 and 3V3





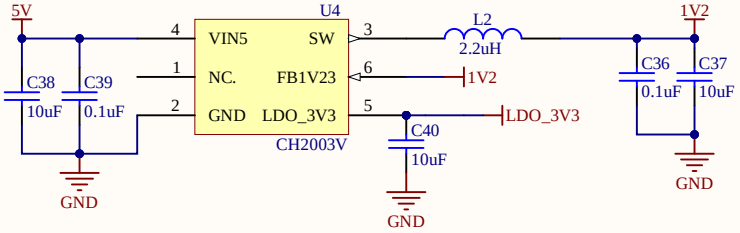


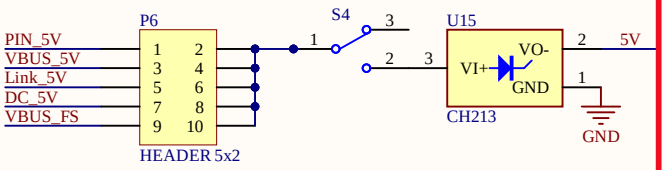




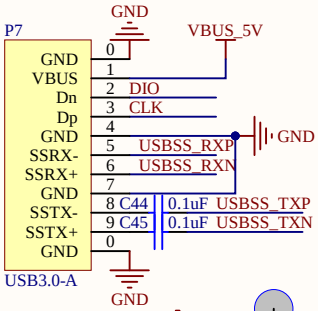


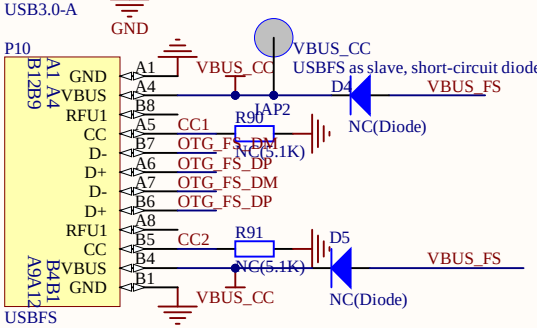
3V3 1V2 NET





USB SUPPLY



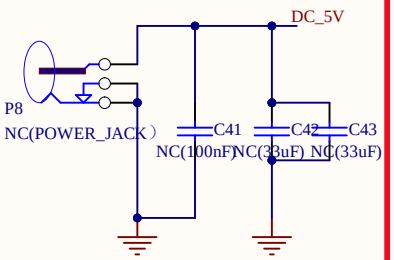


USBFS as slave, short-circuit diode is required; as master, low-dropout ideal diode is required to maintain VBUS_CC 4.7V

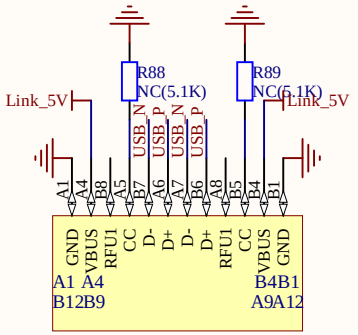


USBPD does not power the MCU, and other methods of powering the system need to be selected additionally when VBUS_CC is induced

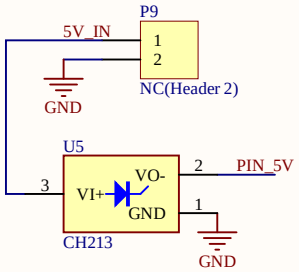
DC SUPPLY



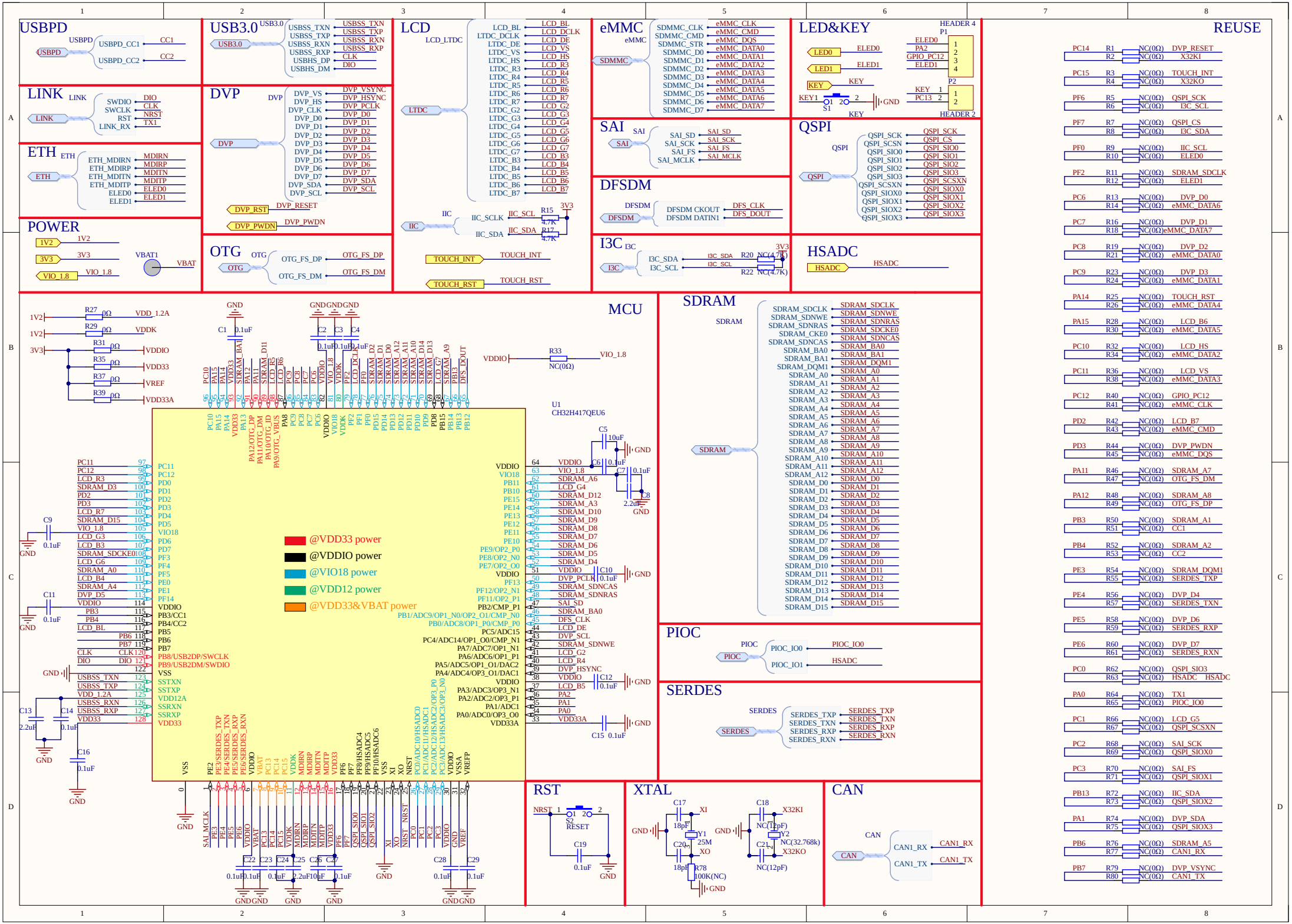
LINK SUPPLY



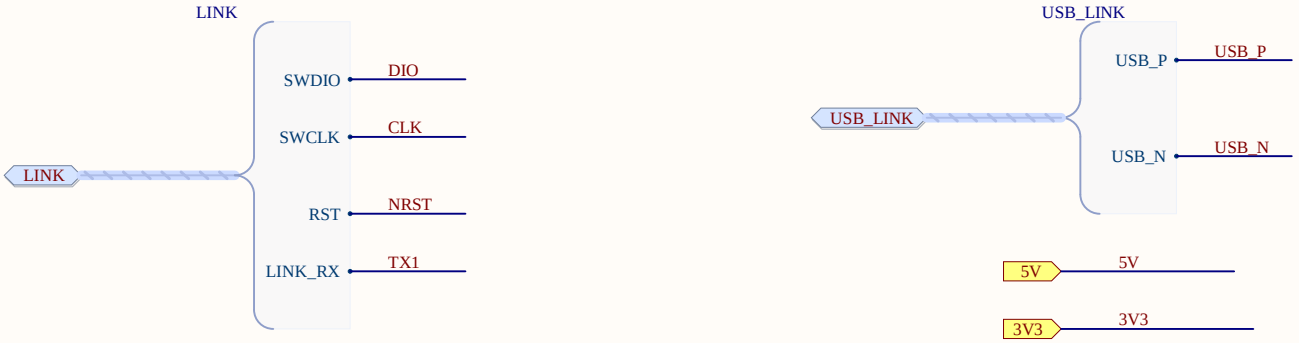
PIN SUPPLY

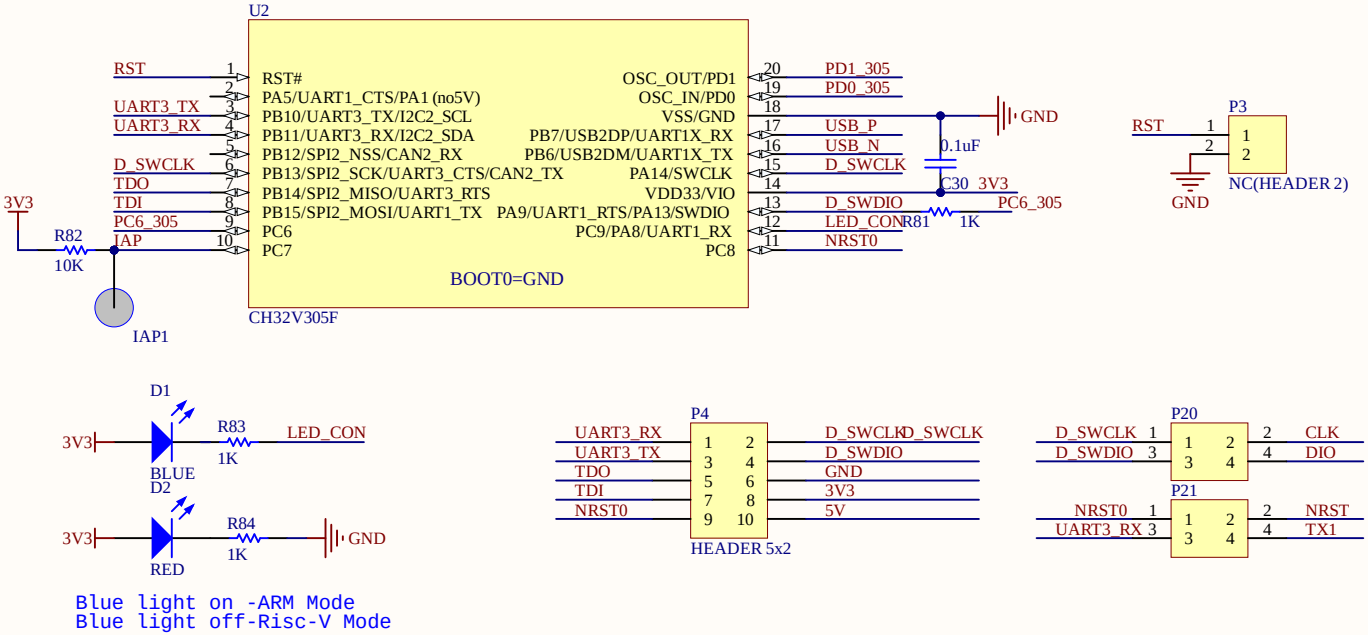


CH213 used to protect against power short circuits


USBPD
USB3.0
LCD
eMMC
LED&KEY
REUSE
LINK
DVP
SAI
QSPI
ETH
DFSDM
POWER
OTG
I3C
HSADC
MCU
SDRAM
PIOC
SERDES
RST
XTAL
CAN
@VDD33 power
@VDDIO power
@VIO18 power
@VDD12 power
@VDD33&VBAT power

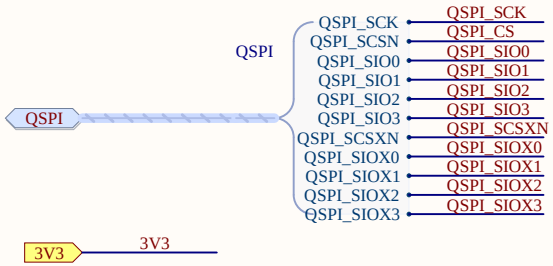
LINK

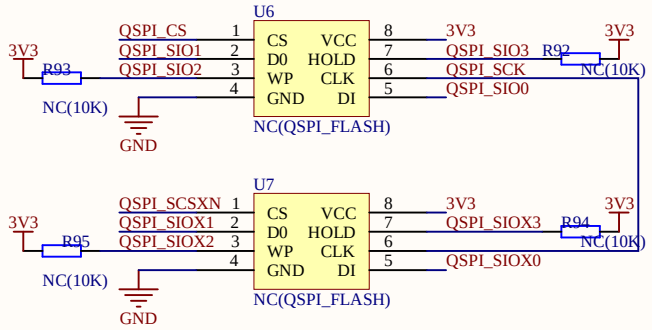




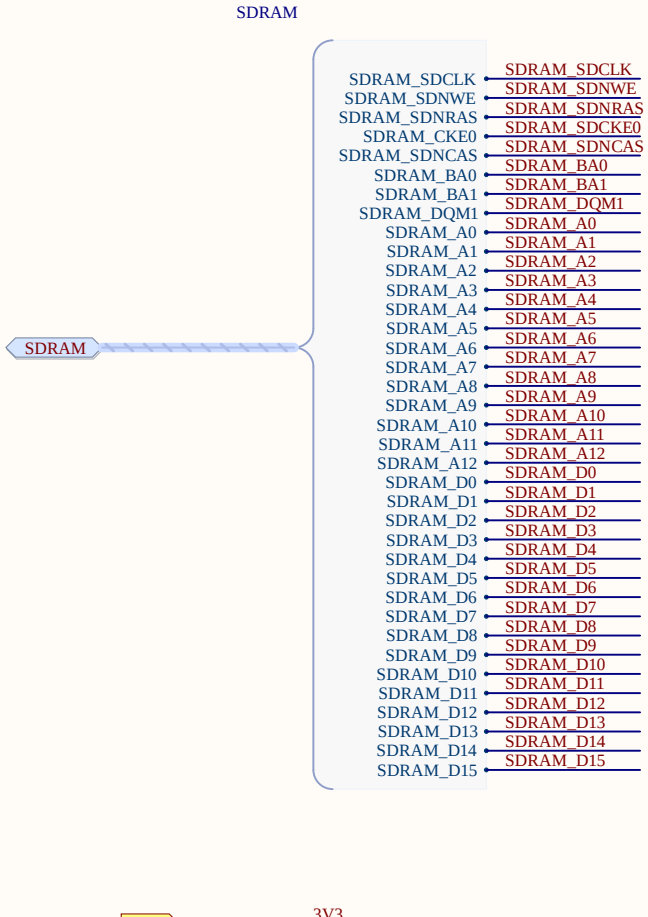
Blue light on -ARM Mode
Blue light off-Risc-V Mode

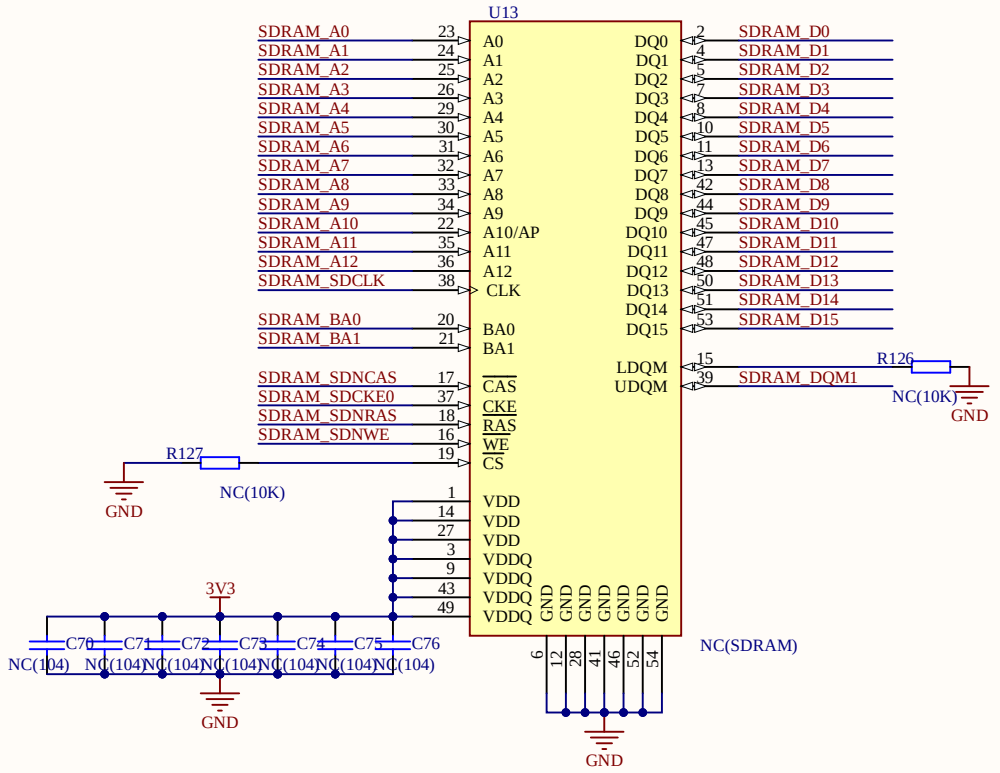
QSPI


QSPI
QSPI_SCK
QSPI_SCSN
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_SCSXN
QSPI_SIOX0
QSPI_SIOX1
QSPI_SIOX2
QSPI_SIOX3
QSPI_SCK
QSPI_CS
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_SCSXN
QSPI_SIOX0
QSPI_SIOX1
QSPI_SIOX2
QSPI_SIOX3
3V3
3V3

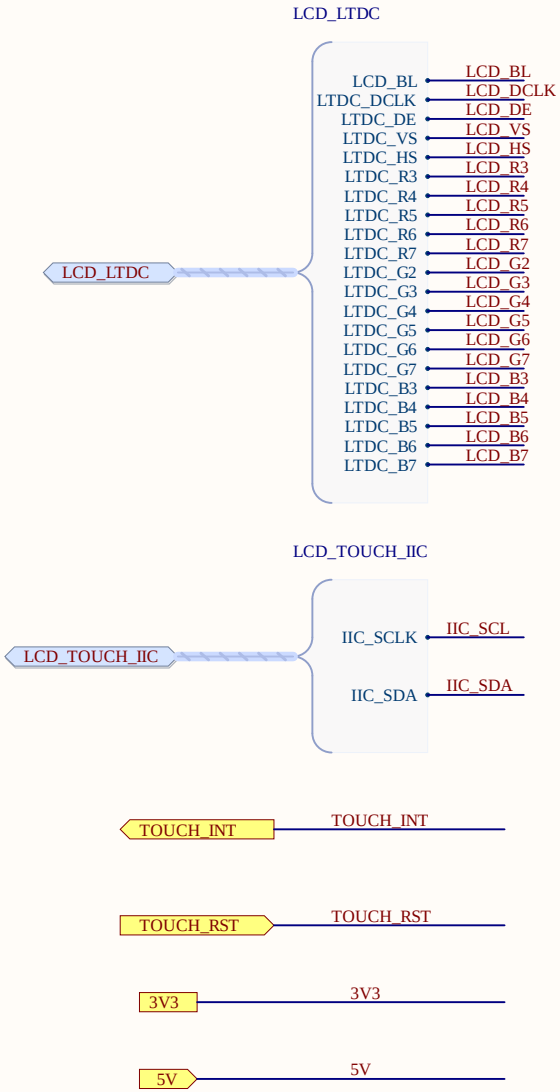

U6
QSPI_CS
QSPI_SIO1
QSPI_SIO2
CS
D0
WP
GND
VCC
HOLD
CLK
DI
3V3
QSPI_SIO3
QSPI_SCK
QSPI_SIO0
3V3
R92
NC(10K)
3V3
R93
NC(10K)
GND
NC(QSPI_FLASH)
U7
QSPI_SCSXN
QSPI_SIOX1
QSPI_SIOX2
3V3
QSPI_SIOX3
QSPI_SIOX0
3V3
R94
NC(10K)
3V3
R95
NC(10K)
GND
NC(QSPI_FLASH)

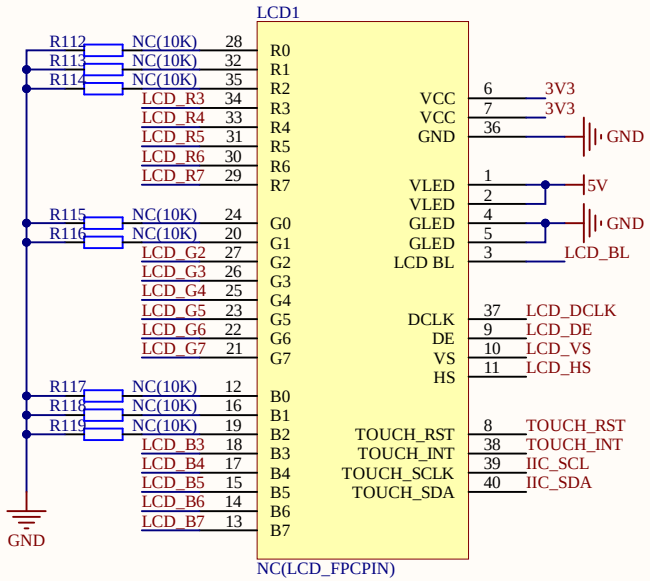
SDRAM


SDRAM
SDRAM
SDRAM_SDCLK
SDRAM_SDNWE
SDRAM_SDNRAS
SDRAM_CKE0
SDRAM_SDNCAS
SDRAM_BA0
SDRAM_BA1
SDRAM_DQM1
SDRAM_A0
SDRAM_A1
SDRAM_A2
SDRAM_A3
SDRAM_A4
SDRAM_A5
SDRAM_A6
SDRAM_A7
SDRAM_A8
SDRAM_A9
SDRAM_A10
SDRAM_A11
SDRAM_A12
SDRAM_D0
SDRAM_D1
SDRAM_D2
SDRAM_D3
SDRAM_D4
SDRAM_D5
SDRAM_D6
SDRAM_D7
SDRAM_D8
SDRAM_D9
SDRAM_D10
SDRAM_D11
SDRAM_D12
SDRAM_D13
SDRAM_D14
SDRAM_D15
3V3
3V3

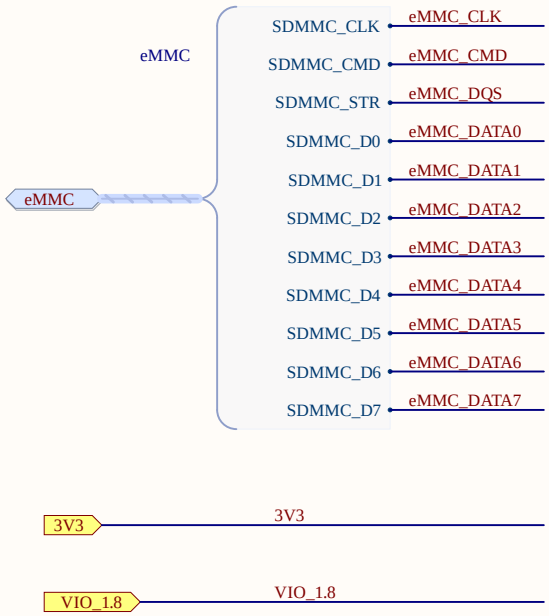

U13
SDRAM_A0 23 A0
SDRAM_A1 24 A1
SDRAM_A2 25 A2
SDRAM_A3 26 A3
SDRAM_A4 29 A4
SDRAM_A5 30 A5
SDRAM_A6 31 A6
SDRAM_A7 32 A7
SDRAM_A8 33 A8
SDRAM_A9 34 A9
SDRAM_A10 22 A10/AP
SDRAM_A11 35 A11
SDRAM_A12 36 A12
SDRAM_SDCLK 38 CLK
SDRAM_BA0 20 BA0
SDRAM_BA1 21 BA1
SDRAM_SDNCAS 17 CAS
SDRAM_SDCKE0 37 CKE
SDRAM_SDNRAS 18 RAS
SDRAM_SDNWE 16 WE
19 CS
1 VDD
14 VDD
27 VDD
3 VDDQ
9 VDDQ
43 VDDQ
49 VDDQ
DQ0 2 SDRAM_D0
DQ1 4 SDRAM_D1
DQ2 5 SDRAM_D2
DQ3 7 SDRAM_D3
DQ4 8 SDRAM_D4
DQ5 10 SDRAM_D5
DQ6 11 SDRAM_D6
DQ7 13 SDRAM_D7
DQ8 42 SDRAM_D8
DQ9 44 SDRAM_D9
DQ10 45 SDRAM_D10
DQ11 47 SDRAM_D11
DQ12 48 SDRAM_D12
DQ13 50 SDRAM_D13
DQ14 51 SDRAM_D14
DQ15 53 SDRAM_D15
LDQM 15
UDQM 39 SDRAM_DQM1
R126
NC(10K)
GND
R127
NC(10K)
GND
3V3
C70 NC(104)
C71 NC(104)
C72 NC(104)
C73 NC(104)
C74 NC(104)
C75 NC(104)
C76 NC(104)
GND
GND GND GND GND GND GND
6 12 28 41 46 52 54
NC(SDRAM)
GND

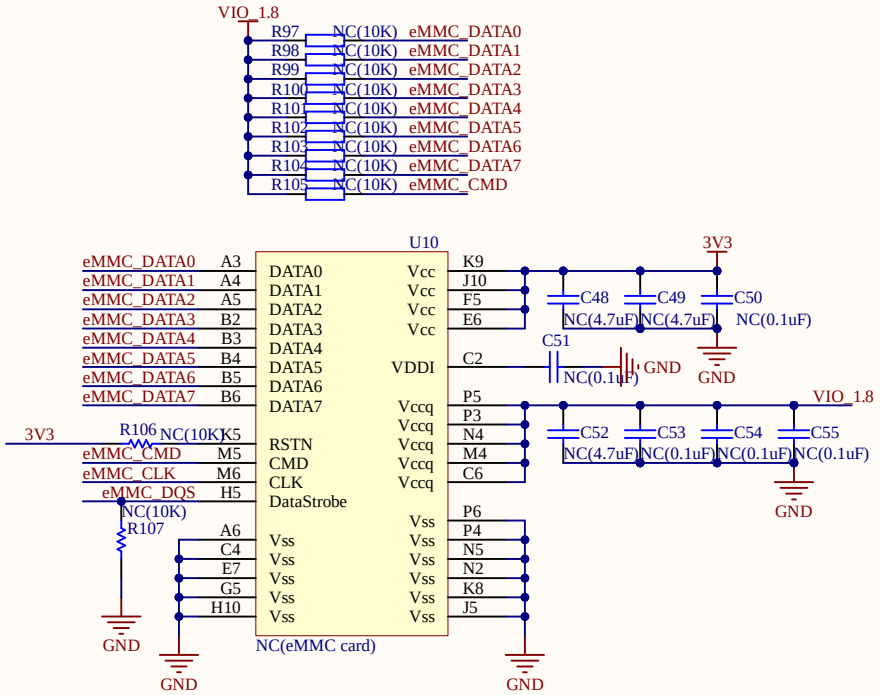
LCD


LCD_LTDC
LCD_BL
LTDC_DCLK
LTDC_DE
LTDC_VS
LTDC_HS
LTDC_R3
LTDC_R4
LTDC_R5
LTDC_R6
LTDC_R7
LTDC_G2
LTDC_G3
LTDC_G4
LTDC_G5
LTDC_G6
LTDC_G7
LTDC_B3
LTDC_B4
LTDC_B5
LTDC_B6
LTDC_B7
LCD_BL
LCD_DCLK
LCD_DE
LCD_VS
LCD_HS
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
LCD_TOUCH_IIC
IIC_SCLK
IIC_SDA
IIC_SCL
IIC_SDA
TOUCH_INT
TOUCH_INT
TOUCH_RST
TOUCH_RST
3V3
3V3
5V
5V

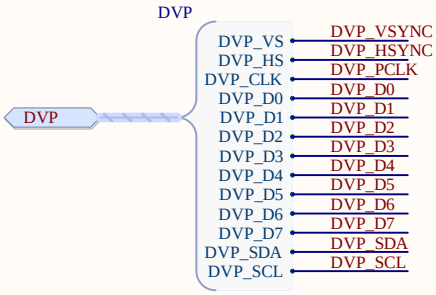

LCD1
R112
R113
R114
R115
R116
R117
R118
R119
NC(10K)
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
R0
R1
R2
R3
R4
R5
R6
R7
G0
G1
G2
G3
G4
G5
G6
G7
B0
B1
B2
B3
B4
B5
B6
B7
VCC
GND
VLED
GLED
LCD BL
DCLK
DE
VS
HS
TOUCH_RST
TOUCH_INT
TOUCH_SCLK
TOUCH_SDA
3V3
5V
LCD_BL
LCD_DCLK
LCD_DE
LCD_VS
LCD_HS
TOUCH_RST
TOUCH_INT
IIC_SCL
IIC_SDA
NC(LCD_FPCPIN)

eMMC


eMMC
SDMMC_CLK
SDMMC_CMD
SDMMC_STR
SDMMC_D0
SDMMC_D1
SDMMC_D2
SDMMC_D3
SDMMC_D4
SDMMC_D5
SDMMC_D6
SDMMC_D7
eMMC_CLK
eMMC_CMD
eMMC_DQS
eMMC_DATA0
eMMC_DATA1
eMMC_DATA2
eMMC_DATA3
eMMC_DATA4
eMMC_DATA5
eMMC_DATA6
eMMC_DATA7
eMMC
3V3
3V3
VIO_1.8
VIO_1.8


VIO_1.8
R97 NC(10K) eMMC_DATA0
R98 NC(10K) eMMC_DATA1
R99 NC(10K) eMMC_DATA2
R100 NC(10K) eMMC_DATA3
R101 NC(10K) eMMC_DATA4
R102 NC(10K) eMMC_DATA5
R103 NC(10K) eMMC_DATA6
R104 NC(10K) eMMC_DATA7
R105 NC(10K) eMMC_CMD
U10
eMMC_DATA0 A3
eMMC_DATA1 A4
eMMC_DATA2 A5
eMMC_DATA3 B2
eMMC_DATA4 B3
eMMC_DATA5 B4
eMMC_DATA6 B5
eMMC_DATA7 B6
3V3
R106 NC(10K)
eMMC_CMD M5
eMMC_CLK M6
eMMC_DQS H5
NC(10K)
R107
GND
A6
C4
E7
G5
H10
GND
DATA0
DATA1
DATA2
DATA3
DATA4
DATA5
DATA6
DATA7
RSTN
CMD
CLK
DataStrobe
Vss
Vss
Vss
Vss
Vss
NC(eMMC card)
Vcc
Vcc
Vcc
Vcc
VDDI
Vccq
Vccq
Vccq
Vccq
Vccq
Vss
Vss
Vss
Vss
Vss
K9
J10
F5
E6
C2
P5
P3
N4
M4
C6
P6
P4
N5
N2
K8
J5
GND
3V3
C48 NC(4.7uF)
C49 NC(4.7uF)
C50 NC(0.1uF)
GND
C51
NC(0.1uF)
GND
VIO_1.8
C52 NC(4.7uF)
C53 NC(0.1uF)
C54 NC(0.1uF)
C55 NC(0.1uF)
GND

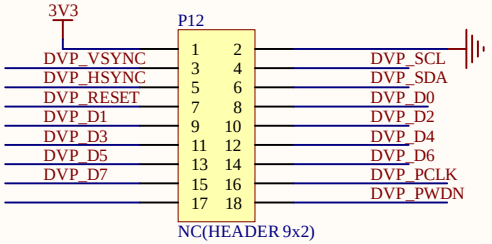
DVP


DVP
DVP
DVP_VS
DVP_HS
DVP_CLK
DVP_D0
DVP_D1
DVP_D2
DVP_D3
DVP_D4
DVP_D5
DVP_D6
DVP_D7
DVP_SDA
DVP_SCL
DVP_VSYNC
DVP_HSYNC
DVP_PCLK
DVP_D0
DVP_D1
DVP_D2
DVP_D3
DVP_D4
DVP_D5
DVP_D6
DVP_D7
DVP_SDA
DVP_SCL

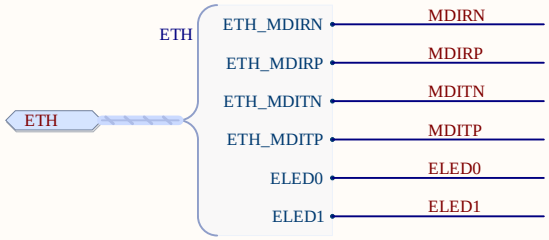

DVP_RST
DVP_RESET


DVP_PWDN
DVP_PWDN


3V3
3V3

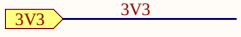

3V3
P12
DVP_VSYNC
DVP_HSYNC
DVP_RESET
DVP_D1
DVP_D3
DVP_D5
DVP_D7
1 2
3 4
5 6
7 8
9 10
11 12
13 14
15 16
17 18
DVP_SCL
DVP_SDA
DVP_D0
DVP_D2
DVP_D4
DVP_D6
DVP_PCLK
DVP_PWDN
NC(HEADER 9x2)

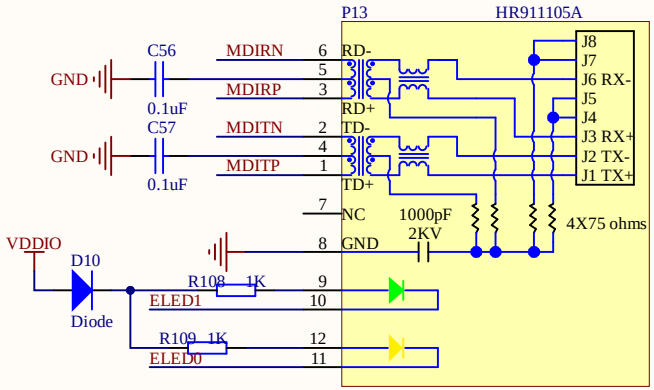
ETH





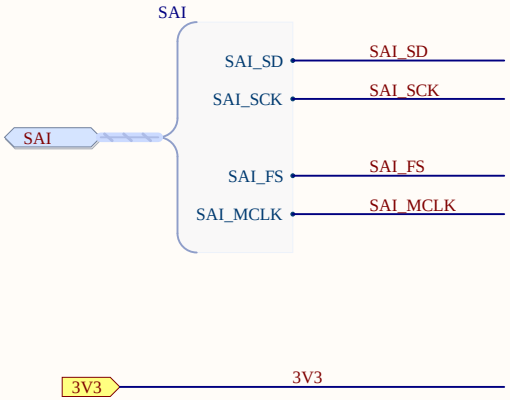


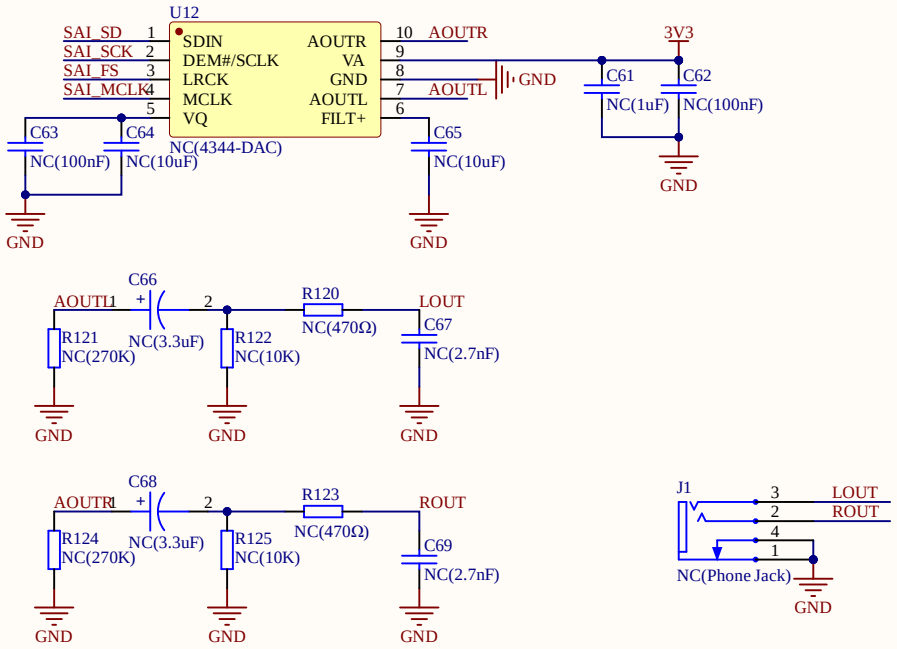




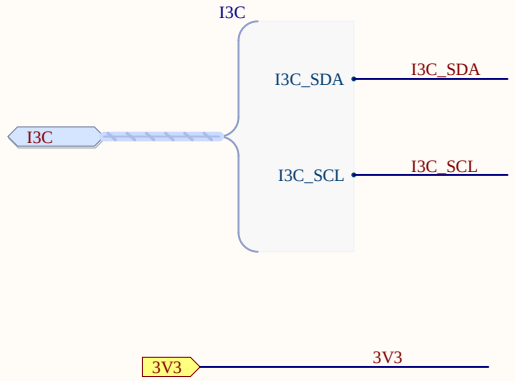
If VIO18 is below 1.8V, the Ethernet port indicator circuit needs to be revised.
The Ethernet port LED cannot work.To prevent current backflow,do not populate resistor R108 and R109.

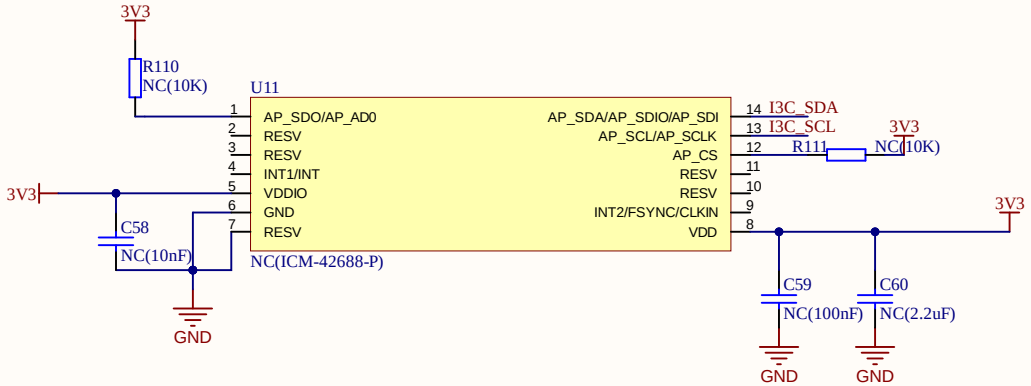
SAI


SAI
SAI_SD
SAI_SCK
SAI_FS
SAI_MCLK
SAI
SAI_SD
SAI_SCK
SAI_FS
SAI_MCLK
3V3
3V3

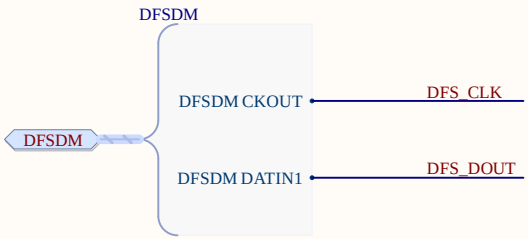

U12
SAI_SD 1
SAI_SCK 2
SAI_FS 3
SAI_MCLK 4
5
SDIN
DEM#/SCLK
LRCK
MCLK
VQ
AOUTR
VA
GND
AOUTL
FILT+
10 AOUTR
9
8
7 AOUTL
6
NC(4344-DAC)
C63
NC(100nF)
C64
NC(10uF)
C65
NC(10uF)
GND
GND
GND
3V3
C61
NC(1uF)
C62
NC(100nF)
GND
AOUTL 1
C66
NC(3.3uF)
2
R120
NC(470Ω)
LOUT
C67
NC(2.7nF)
R121
NC(270K)
R122
NC(10K)
GND
GND
GND
AOUTR 1
C68
NC(3.3uF)
2
R123
NC(470Ω)
ROUT
C69
NC(2.7nF)
R124
NC(270K)
R125
NC(10K)
GND
GND
GND
J1
3
2
4
1
LOUT
ROUT
NC(Phone Jack)
GND

I3C

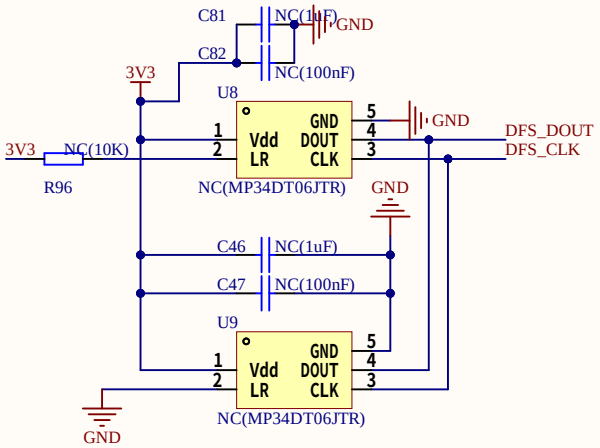

I3C
I3C_SDA
I3C_SCL
I3C
I3C_SDA
I3C_SCL
3V3
3V3


3V3
R110
NC(10K)
U11
AP_SDO/AP_AD0
RESV
RESV
INT1/INT
VDDIO
GND
RESV
AP_SDA/AP_SDIO/AP_SDI
AP_SCL/AP_SCLK
AP_CS
RESV
RESV
INT2/FSYNC/CLKIN
VDD
NC(ICM-42688-P)
3V3
C58
NC(10nF)
GND
I3C_SDA
I3C_SCL
3V3
R111
NC(10K)
3V3
C59
NC(100nF)
C60
NC(2.2uF)
GND
GND

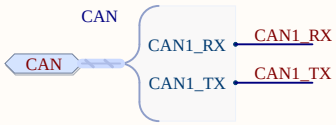
DFSDM


DFSDM
DFSDM
DFSDM CKOUT
DFSDM DATIN1
DFS_CLK
DFS_DOUT

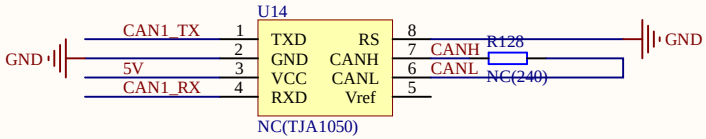

3V3
3V3

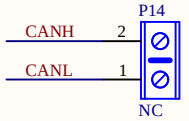

C81
NC(1uF)
GND
C82
NC(100nF)
3V3
U8
GND
5
4
3
1
2
Vdd
DOUT
LR
CLK
GND
DFS_DOUT
DFS_CLK
3V3
NC(10K)
R96
NC(MP34DT06JTR)
GND
C46
NC(1uF)
C47
NC(100nF)
U9
GND
5
4
3
1
2
Vdd
DOUT
LR
CLK
NC(MP34DT06JTR)
GND

CAN


CAN
CAN
CAN1_RX
CAN1_TX
CAN1_RX
CAN1_TX

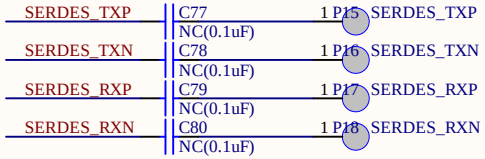

5V
5V


U14
GND
CAN1_TX
5V
CAN1_RX
1
2
3
4
TXD
GND
VCC
RXD
RS
CANH
CANL
Vref
8
7
6
5
CANH
CANL
R128
NC(240)
GND
NC(TJA1050)


P14
CANH
CANL
2
1
NC

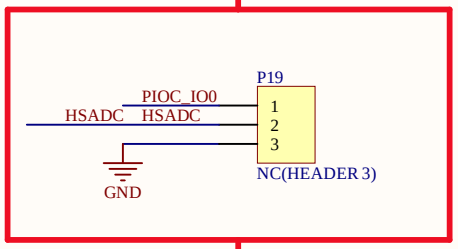
SERDES





HSADC





PIOC

